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Department of Health Research
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Standard Treatment Workflow (STW)

SEPSIS IN NEONATES

ICD-10-P36

Assess every neonate born in or brought to a health facility for presence of sepsis, at admission and during hospital stay, by looking for red and yellow flag signs and risk factors



RED FLAGS

- Shock
- Bleeding from multiple sites
- Sclerema – hardened skin over cheeks / thighs
- Respiratory distress needing intubation or Silverman's score >5
- If age of baby is less than 72 hours and mother has foul-smelling liquor or chorioamnionitis

YELLOW FLAGS: SIGNS OR MATERNAL RISK FACTORS

Seizures*	Refusal to feed	HR >180 persisting for one hour despite normal temperature	Fast breathing (RR >60/min)	Floppiness
Lethargy	Feed intolerance	New or increased apneic episodes	Fever, or hypothermia <36.0°C not corrected by rewarming	

Any of the maternal risk factors: If age of baby is less than 3 days (72 hrs) and mother has

>3 PV examinations during labour	Rupture of membranes >24 hrs	pPROM	Urinary tract infection	Diarrhea	Fever >38°C

HIGH PROBABILITY OF SEPSIS

START TREATMENT AND INVESTIGATE

- Any RED flag is present
- Any two YELLOW flags (signs or maternal risk factors) are present
- One YELLOW flag (sign or maternal risk factor) is present AND baby's gestation at birth is ≤ 32 weeks or birth weight <1500 g

- Admit in the NICU / SNCU
- Obtain blood sample for culture and sensitivity (even if baby has received an antibiotic before admission)
- Start empirical antibiotics as per local/unit policy, pending reports
- Provide supportive care: thermal care, fluids & feeds, glucose monitoring, respiratory & circulatory support
- Do labs as indicated (X-ray, CBC, platelets, RBS, electrolytes, renal function)
- Perform lumbar puncture (LP) for CSF analysis when baby is hemodynamically stable

AT-RISK / SUSPECT SEPSIS

OBSERVE

- One YELLOW flag (sign or maternal risk factor) is present AND
- Baby's gestation at birth is >32 weeks

- Keep under close serial observation for 48–72 hrs (monitor 4–6 hourly)
- Start antibiotics if another yellow / red sign appears during observation
- Obtain sample for blood culture and sensitivity before starting antibiotics (even if baby has received an antibiotic before admission)
- Perform LP for CSF analysis if starting antibiotics or if the blood culture is positive

* Neonatal seizure always warrants urgent evaluation for cause (sepsis / meningitis, hypoglycaemia). Keep a low threshold to start antibiotics and perform LP.

REVIEW AT 48 HRS

SIGNS OF SEPSIS DISAPPEARED

- Stop antibiotics
- Keep under observation till blood culture is reported as sterile after 48 hrs of incubation

SIGNS OF SEPSIS IMPROVING BUT STILL PRESENT

- Continue antibiotics
- Antibiotic duration based on blood culture and LP report

SIGNS OF SEPSIS WORSENED, OR A RED FLAG APPEARED AFTER STARTING TREATMENT

- Upgrade antibiotics as per antibiotic local/ unit policy
- Antibiotic duration based on blood culture and LP report

If antibiotics were continued, review at 5 days. If the baby is now well, the 48-h blood culture is sterile and CSF is normal, stop antibiotics.

A sterile blood culture at 24–48 h after starting antibiotics supports early stopping of antibiotics.

DURATION OF ANTIBIOTICS

CONDITION	DURATION
Pneumonia	**5–7 DAYS**
Clinically suspected sepsis AND sterile blood culture AND normal CSF analysis	**5–7 DAYS**
Blood culture positive (if asymptomatic by day 5, no meningitis)	**7–10 DAYS**
Blood culture positive (if remains symptomatic after day 5, no meningitis)	**10–14 DAYS**
CSF suggestive of meningitis	**21 DAYS**

DOs

- Start antibiotics within 1 hour if high probability of sepsis is present
- Draw blood culture before the first antibiotic dose
- Review antibiotics at 48 h against culture, CRP and clinical course
- Stop antibiotics if blood culture is sterile and the baby has recovered
- Use serial observation and look for non-infective causes of illness

DON'Ts

- Start antibiotics without an indication – clinical signs in neonates are non-specific
- Continue antibiotics for a sterile culture in a baby who has recovered
- Act on a positive CRP / procalcitonin in isolation – they are false positive in many non-infective conditions; use to rule out sepsis
- Delay LP when it is indicated once the baby is stable

KEY PERFORMANCE INDICATORS (KPIs)

Blood culture before first antibiotic dose (%)

$$\frac{\text{Sepsis episodes with blood culture drawn before the first antibiotic dose}}{\text{Sepsis episodes in which antibiotics were started}} \times 100$$

Antibiotic initiation within 1 hour (%)

$$\frac{\text{Neonates with high probability of sepsis who received the first antibiotic dose within 1 hour of identification}}{\text{All neonates identified as having high probability of sepsis}} \times 100$$

Antibiotic exposure during admission (%)

$$\frac{\text{Neonates admitted during the reporting period who received at least one dose of a systemic antibiotic during their stay}}{\text{All neonates admitted during the same reporting period}} \times 100$$

PREVENT SEPSIS BY ENSURING HAND HYGIENE, ASEPSIS DURING PROCEDURES AND DILIGENT HOUSEKEEPING

RELATED INFORMATION

Hand hygiene
youtu.be/lkp5BXzPYBA

Blood culture
youtu.be/6pQBzt-EZN8

Lumbar puncture
youtu.be/6-h4xUIrMCg

ABBREVIATIONS

CBC: Complete blood count
CRP: C-reactive protein
CSF: Cerebrospinal fluid
LP: Lumbar puncture
NICU: Neonatal intensive care unit
pPROM: Preterm premature rupture of membranes
RBS: Random blood sugar
SAS: Silverman-Andersen score
SNCU: Special newborn care unit

REFERENCES

1. The Young Infants Clinical Signs Study Group. Clinical signs that predict severe illness in children under age 2 months: a multicentre study. Lancet. 2008;371(9607):135–42.
2. Neonatal infection: antibiotics for prevention and treatment. London: National Institute for Health and Care Excellence (NICE); 2021 Apr 20. (NICE Guideline, No.195). Available from: https://www.nice.org.uk/guidance/ng195 Last access on 20/04/2022

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